A Lightweight Canonicalization Engine in C for Disciplined Convex Programming

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Outline

Introduction

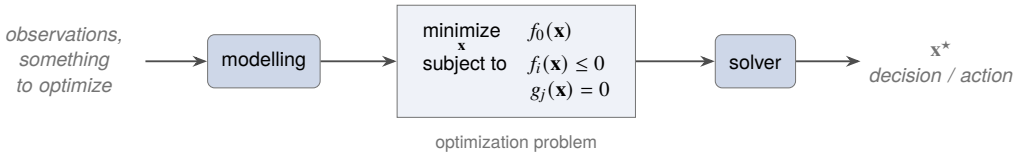
Background

Single canonicalization

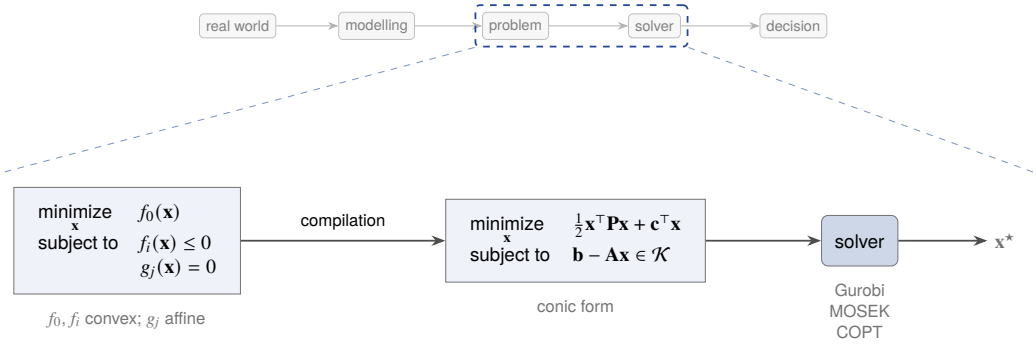
Parametric canonicalization

Results

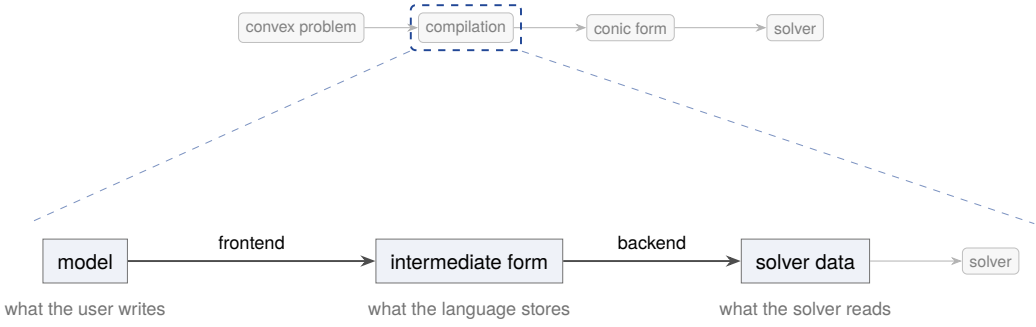
Optimization in practice



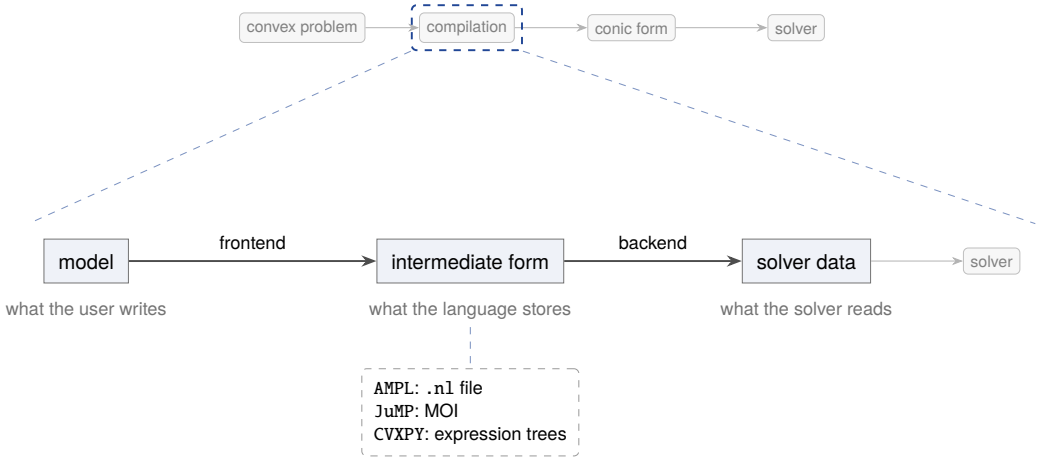
Convex optimization



Compilation



Compilation

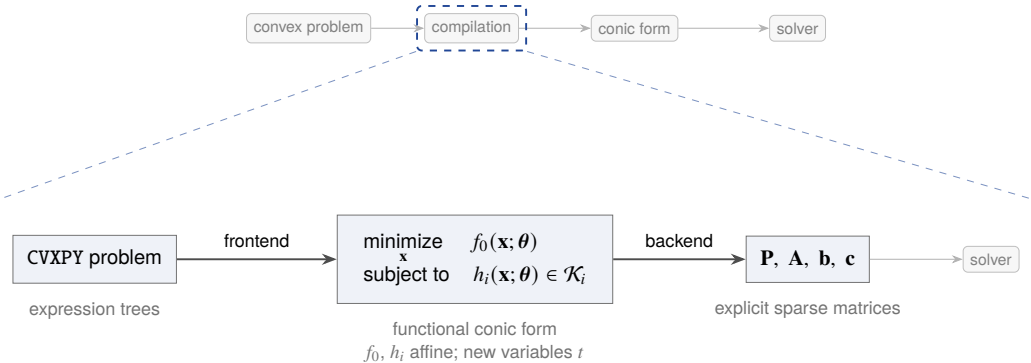


Disciplined convex programming

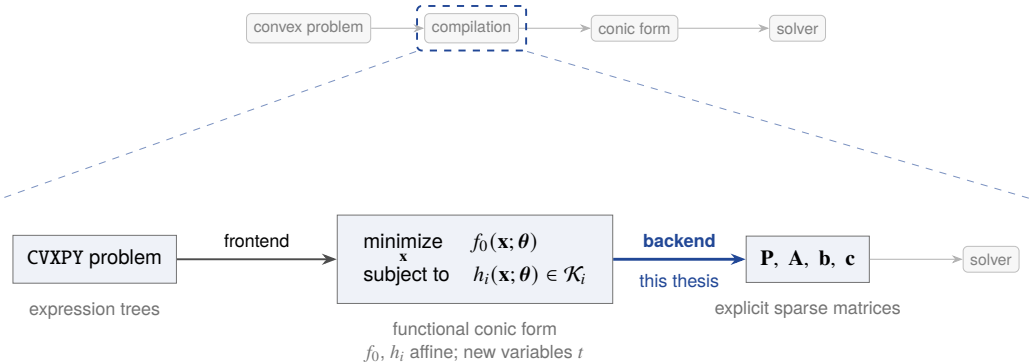
DCP-based modelling languages

Atoms and expression trees

Canonicalization: frontend and backend



Canonicalization: frontend and backend



Problem statement

Outline

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Single canonicalization

Parametric canonicalization

Results

Conic reformulation: epigraphs and cones

Quadratic conic standard form

Derivatives: algorithmic differentiation and graph colouring

Dense and sparse matrices

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Thank you

Backup:

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